

Student's Weekly Progress Report

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Course: Bachelor of Science in Computer Science

Report Date	Date Submitted	Task Completed	Task in Progress	Next Week's Task	Problems / Challenges	Field Supervisor Comments
12-06-2026	14-06-2026	• Read project materials and carried out background research to understand the project, especially the use of sound signals in detecting bee behaviors. • Learned how to clean audio files by identifying and removing unwanted noise such as birds, strong wind, knocking sounds on the hive, and other background disturbances. • Gained practical knowledge on how to use audio tools, especially Audacity and Adobe Audition, for waveform and spectrum viewing, noise select	• Clean more bee audio files at a faster pace because I am now getting used to Audacity and Adobe Audition. • Continue practicing noise reduction, amplitude reduction, and selection of useful audio segments.	• Carry out more research on how cleaned audio files can be used in code for feature extraction and machine learning. • Begin connecting the cleaned audio workflow to the coding stage by studying how to load audio files, extract features, and prepare them for analysis. • Clean more bee audio files at a faster pace because I am now getting used to Audacity and Adobe Audition.	• At the beginning, it took time to understand the tools and how to adjust audio views, especially when switching between time, waveform, and frequency/spectrum views. • Some noises were difficult to remove because they overlapped with important bee sounds. This made it necessary to clean carefully so that useful information was not destroyed. • Working with audio requires patience because every recording may contain different types of background noise.	he is on track and is always seeking to understand more about the subject he is assigned to

19-06-2026	19-06-2026	Completed audio cleaning for the bee sound data to prepare it for feature extraction and model testing. Designed a questionnaire to collect field data from the field, which will support project documentation. In addition, I developed the project workflow logic and flow diagram showing how the system reads temperature, humidity, CO₂, and sound data to detect possible signs of honey production or honey processing activity.	I am currently working on designing demographs to show how temperature, humidity, CO₂, and sound values change over time, preparing a sensor position diagram to show where the sensors will be placed inside the hive. In addition, and reviewing related bee hive monitoring projects and preparing demo data to test whether the proposed system logic can work.	- Design demographs to show how the system reads and interprets temperature, humidity, CO₂, and sound data over time. - Create a sensor position diagram to show where sensors will be placed inside the hive and how data will be collected. - Review related projects on bee hive monitoring and honey production monitoring for comparison with my project.	- Reading and reviewing publications and existing research to identify suitable threshold values for temperature, humidity, CO₂, and sound. While some studies provide useful reference ranges, I still need to compare findings from different sources and determine which values are most relevant to my project. - Limited access to some publications due to payment requirements hindered the research process.	Hardworking and innovative individual
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